2D Two-Target Double-Peak Conditions

From Markov-Chain Derivation to Reflecting-Boundary Heatmap Validation

Executive Summary

This report addresses one question: **under two absorbing targets in 2D, when does the total first-passage distribution show a visible double peak?**

Under reflecting boundaries and local-bias corridor design, the representative cases C1–C4 all show two modes in the total first-passage curve. The mechanism is:

1. exact splitting decomposition of total first-passage, and
2. sufficient mode separation with non-extreme splitting weights.

1 Mathematical Mechanism (From First Principles)

1.1 Single-step random walk definition

On $\Omega = \{0, \ldots, N-1\}^2$ with $N = 31$, the unbiased interior transition rule is

$$P((x, y) \to (x \pm 1, y)) = \frac{q}{4}, \tag{1}$$

$$P((x, y) \to (x, y \pm 1)) = \frac{q}{4}, \tag{2}$$

$$P((x, y) \to (x, y)) = 1 - q. \tag{3}$$

We use $q = 0.2$, hence each move direction is 0.05 and stay probability is 0.8.

Reflecting boundary means out-of-domain move mass is added back to stay probability.

Local-bias rule At a biased site, a fraction $\delta = 0.2$ of the current stay probability is shifted to one arrow direction:

$$\Delta = \delta p_{\text{stay}}, \quad p_{\text{stay}} \leftarrow p_{\text{stay}} - \Delta, \quad p_{\text{arrow}} \leftarrow p_{\text{arrow}} + \Delta. \tag{4}$$

For this setup, arrow-direction probability increases from 0.05 to 0.21, with effective drift

$$v_{\text{eff}} \approx p_{\text{forward}} - p_{\text{backward}} = 0.21 - 0.05 = 0.16 \text{ cells/step}. \tag{5}$$

1.2 Two-target absorbing Markov-chain block form

Let m_1, m_2 be absorbing targets and $\mathcal{T}$ be transient states. With row-vector convention:

$$P = \begin{bmatrix} Q & R \\ 0 & I_2 \end{bmatrix}, \quad R = [r_1 \; r_2]. \tag{6}$$

Initial distribution is α (one-hot at n_0), and transient mass evolves as

$$u_0 = \alpha, \qquad u_t = \alpha Q^t. \tag{7}$$

1.3 Exact splitting decomposition

The splitting first-passage distributions are

$$f_1(t) = F_{n_0 \to (m_1|m_2)}(t) = \alpha Q^{t-1} r_1, \quad f_2(t) = F_{n_0 \to (m_2|m_1)}(t) = \alpha Q^{t-1} r_2. \tag{8}$$

Therefore total first-passage is exactly

$$F_{n_0 \to (m_1;m_2)}(t) = f_1(t) + f_2(t) = \alpha Q^{t-1}(r_1 + r_2). \tag{9}$$

Survival and hazard:

$$S(t) = \alpha Q^t \mathbf{1}, \qquad h(t) = \frac{F_{n_0 \to (m_1;m_2)}(t)}{S(t-1)} = \frac{f_1(t)}{S(t-1)} + \frac{f_2(t)}{S(t-1)}. \tag{10}$$

Valleys occur when both channel contributions are simultaneously low.

1.4 Generating-function form and practical criteria

With $\tilde{f}(z) = \sum_{t \geq 1} f(t) z^t$:

$$\tilde{f}_i(z) = z\,\alpha(I - zQ)^{-1} r_i, \qquad \tilde{F}_{\text{any}}(z) = \tilde{f}_1(z) + \tilde{f}_2(z). \tag{11}$$

At $z = 1$, splitting masses are

$$p_i = \sum_{t \geq 1} f_i(t) = \alpha(I - Q)^{-1} r_i, \qquad p_1 + p_2 = 1. \tag{12}$$

In practice, visible double-peak behavior in $F_{\text{any}}(t)$ needs:

- sufficient mode separation, and
- non-negligible splitting weights for both channels.

2 Model Setup and Real Heatmaps (Not Illustrative Templates)

Fixed (1-based) coordinates: $n_0 = (15, 15)$, $m_1 = (22, 15)$, $m_2 = (7, 7)$. Fast centerline: $(15, 15) \to (22, 15)$. Slow centerline: $(15, 15) \to (15, 27) \to (3, 27) \to (3, 7) \to (7, 7)$.

The yellow/purple corridor cells are biased cells. In the compact overview figures, red arrows are centerline-sampled for readability. The appendix provides full-arrow maps with one arrow for every biased cell.

3 Main Results: Total FPT, Splitting, and Hazard

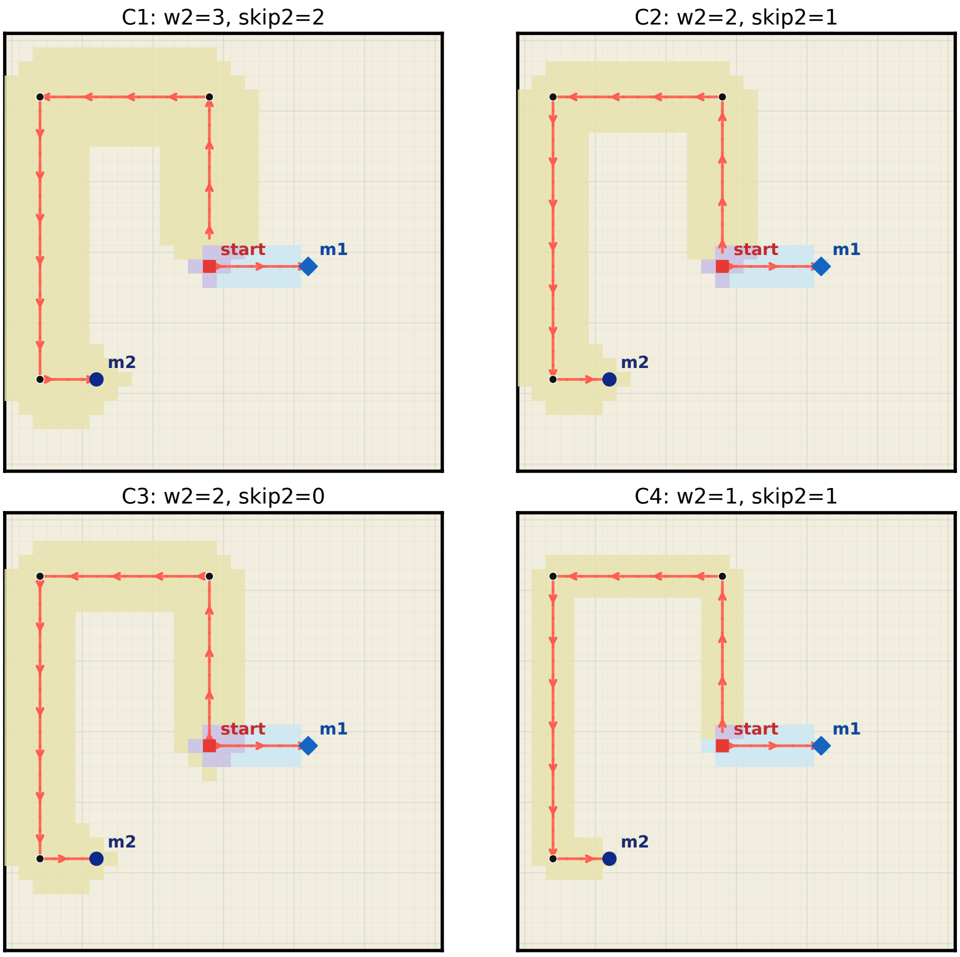

Figure 1: Geometry overview for C1–C4: reflecting lattice background, fast/slow corridor cells, and transport-direction arrows.

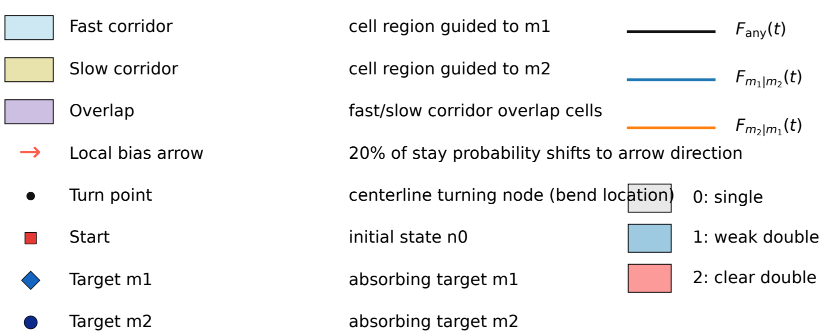

Figure 2: Unified symbol legend panel generated by the same plotting pipeline.

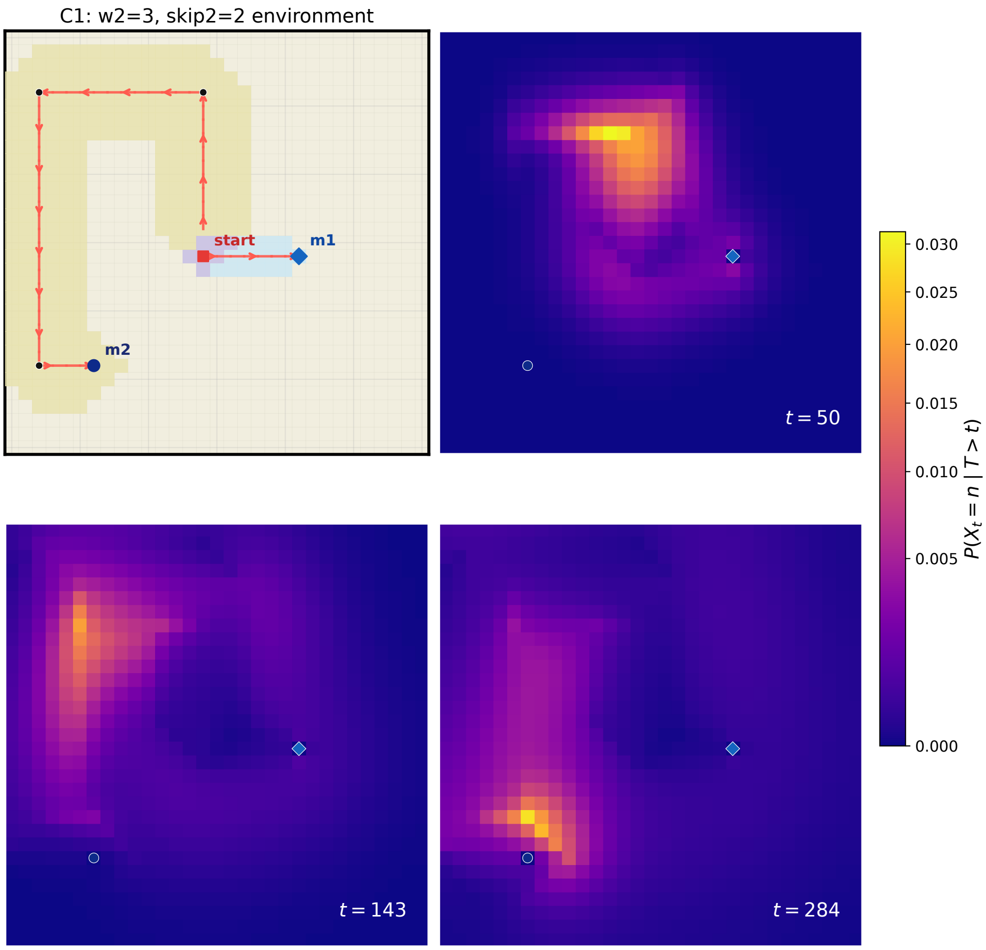

Figure 3: C1 environment + three computed conditional occupancy heatmaps ($t = 50, 143, 284$).

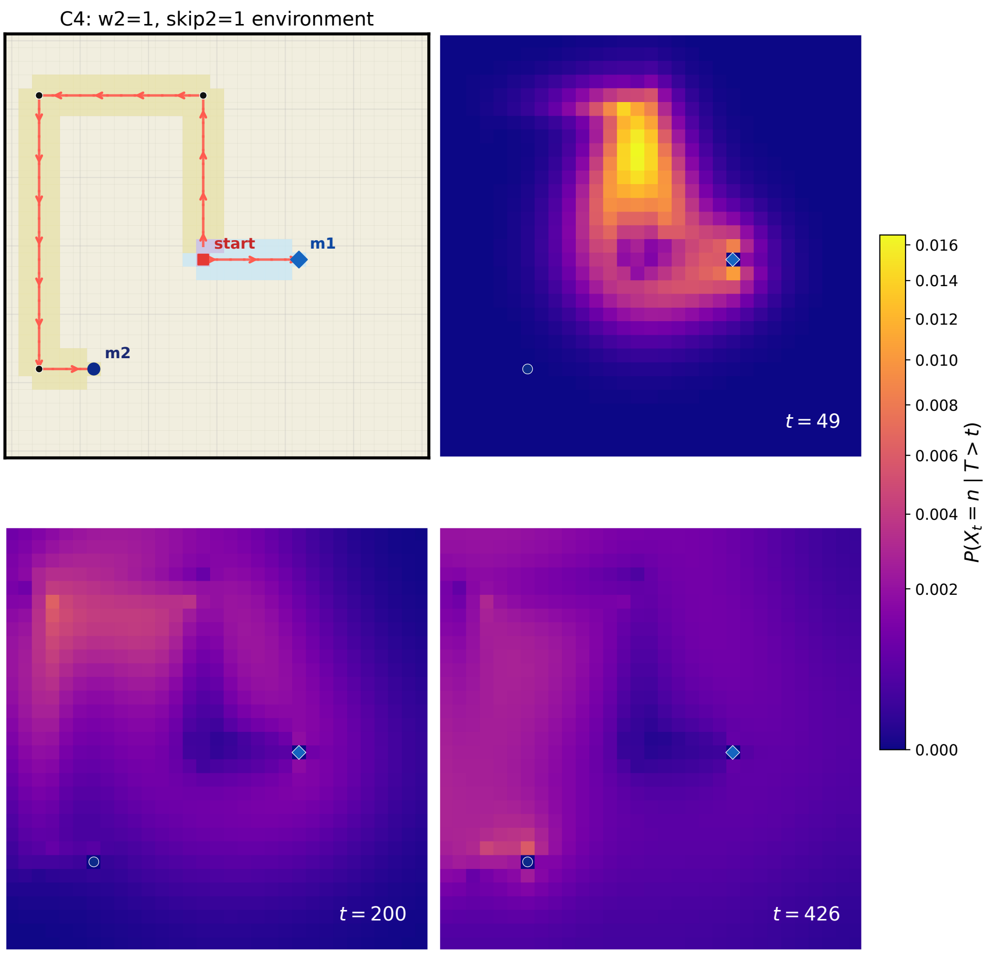

Figure 4: C4 environment + three computed conditional occupancy heatmaps ($t = 49, 200, 426$).

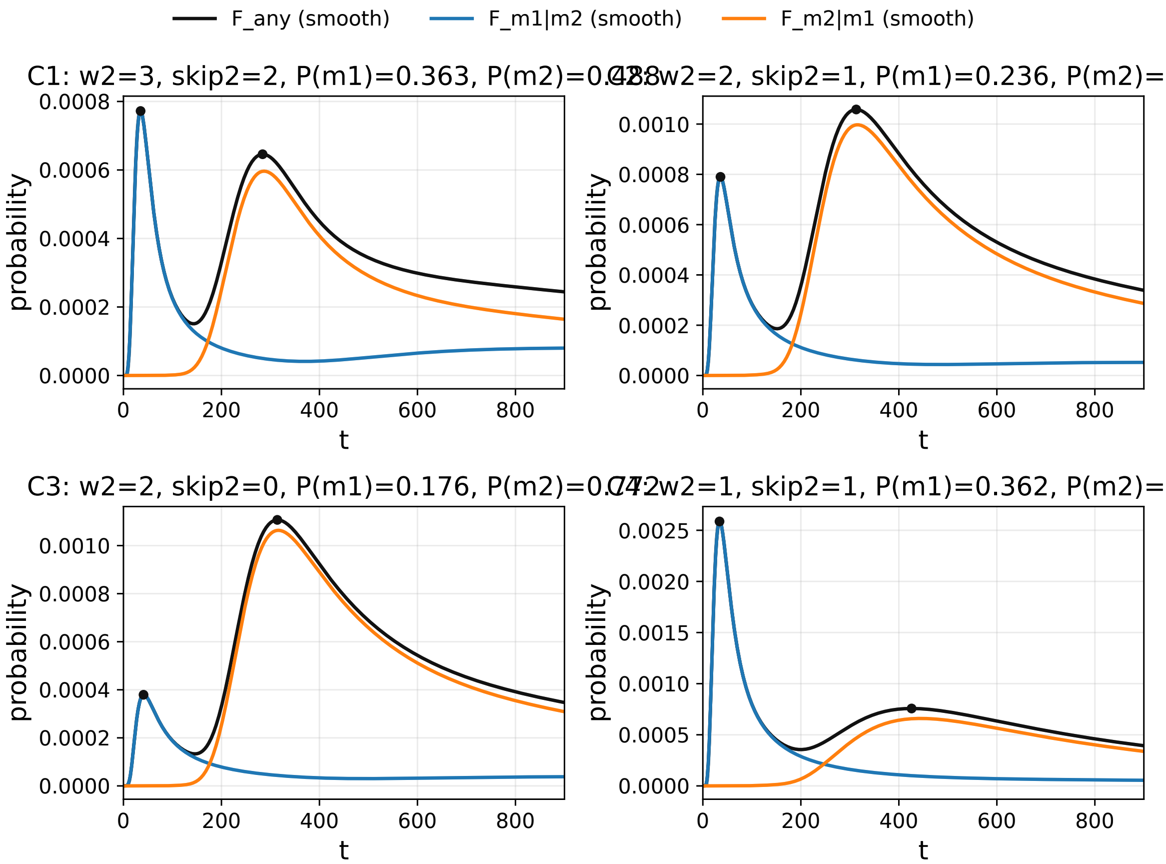

Figure 5: Total first-passage and splitting decomposition for C1–C4. Black: $F_{n_0 \to (m_1;m_2)}(t)$; blue/orange: two splitting components.

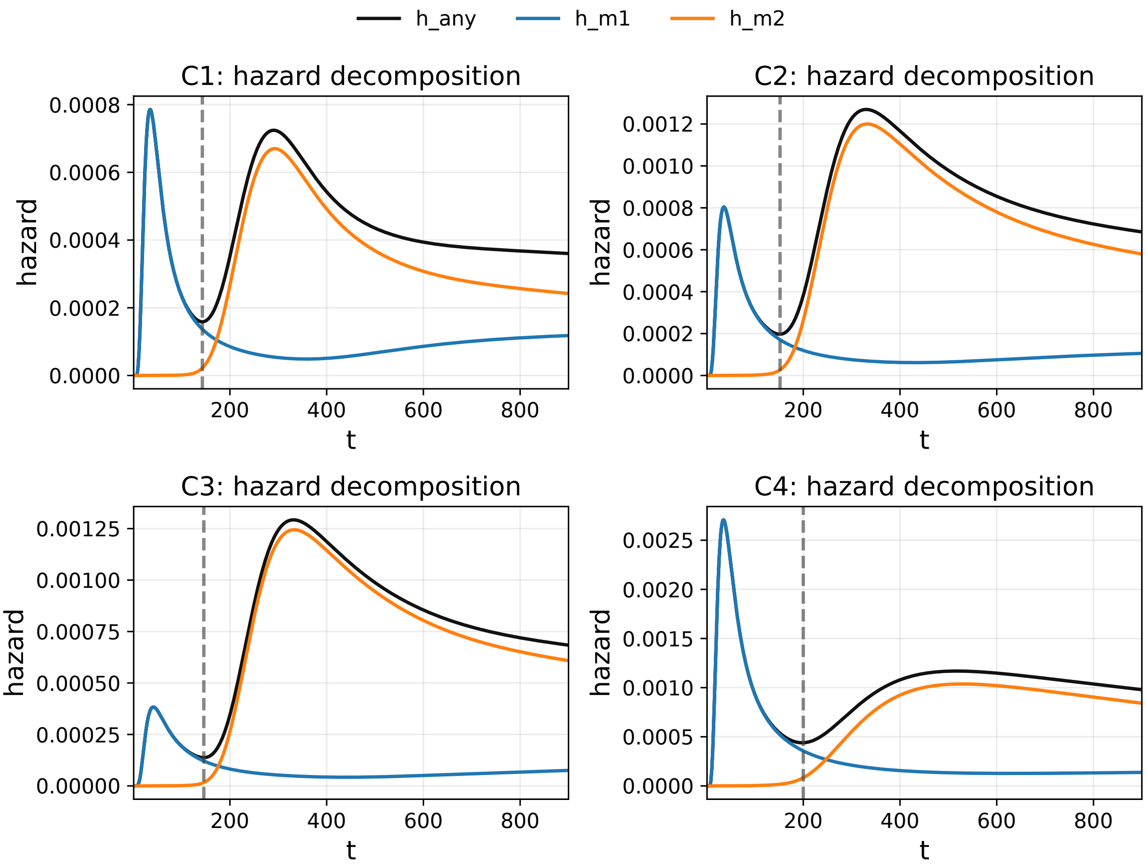

Figure 6: Hazard decomposition $h = h_1 + h_2$. Black: total hazard; blue/orange: channel contributions; dashed line: valley time.

Case	w_2	skip2	t_{p1}	t_{p2}	h_1	h_2	$P(m_1)$	$P(m_2)$
C1	3	2	35	284	7.77e-04	6.46e-04	0.363	0.488
C2	2	1	36	313	7.94e-04	1.06e-03	0.236	0.713
C3	2	0	41	314	3.81e-04	1.11e-03	0.176	0.772
C4	1	1	34	426	2.60e-03	7.57e-04	0.362	0.611

Table 1: Peak locations, peak heights, and splitting weights (with $t_{\max} = 6000$ truncation).

Case	$t^*_{m_1\mid m_2}$	$t^*_{m_2\mid m_1}$	t_v	valley/max	absorbed@6000	tail@6000
C1	35	287	143	0.194	0.851	0.149
C2	36	316	152	0.176	0.949	0.051
C3	41	316	146	0.120	0.948	0.052
C4	34	445	200	0.137	0.973	0.027

Table 2: Splitting modal times, valley depth, and absorbed mass at truncation.

Case	$t^*_{m_1\mid m_2}$	$t^*_{m_2\mid m_1}$	$w^{(1)}_{1/2}$	$w^{(2)}_{1/2}$	$\lvert\Delta t^*\rvert/(w^{(1)}_{1/2} + w^{(2)}_{1/2})$
C1	35	287	25.0	141.5	1.514
C2	36	316	28.5	179.5	1.346
C3	41	316	37.5	177.0	1.282
C4	34	445	26.0	319.0	1.191

Table 3: Mode-separation score based on mode-distance over summed half-widths.

4 Phase Map on $(w_2, \texttt{skip2})$

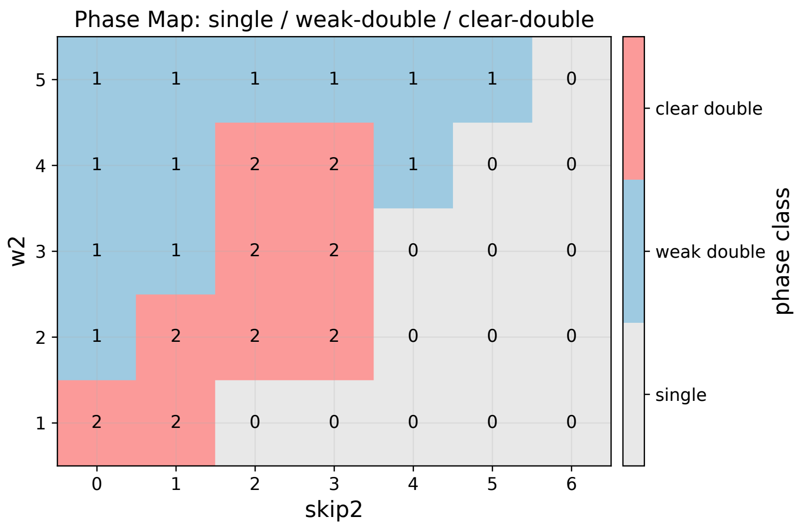

Figure 7: Phase-category map on $(w_2, \texttt{skip2})$: 0=single, 1=weak double, 2=clear double.

5 External Sparse Testset (Double-Peak Cases Only)

Using `external/sparse_double_peak_testset.json`, all 11 sparse-disorder cases were simulated with exact recursion up to $t_{\max} = 6000$. Only phase> 0 cases are retained in this section.

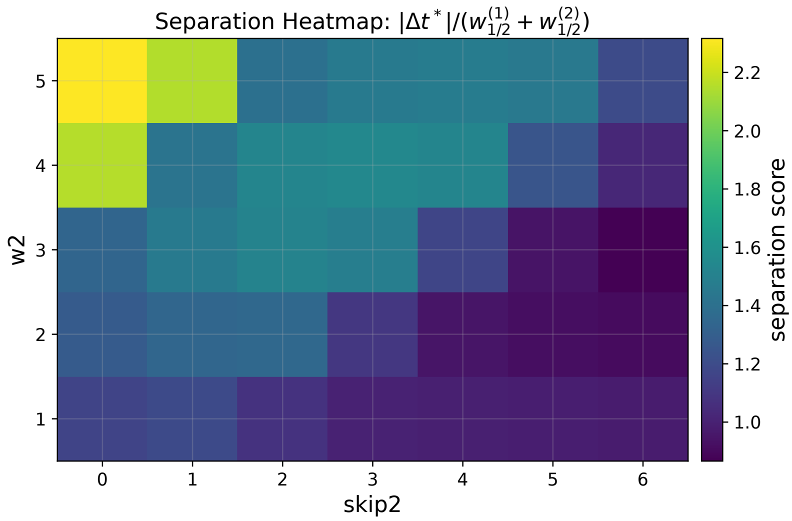

Figure 8: Separation heatmap: $|\Delta t^*|/(w_{1/2}^{(1)} + w_{1/2}^{(2)})$. Brighter means stronger mode separation relative to width.

ID	Name	Type	local-bias	sticky	barriers	shortcuts
S02	corridor thin w2 1 skip2 0	corridor_model	160	0	0	0
S03	centerline only bias fast and slow	explicit_bias_list	54	0	0	0

Table 4: Configuration statistics for retained sparse-testset double-peak cases.

ID	t_{p1}	t_{p2}	valley/max	$P(m_1)$	$P(m_2)$	sep-score	phase
S02	38	436	0.235	0.233	0.740	1.17	2
S03	38	925	0.217	0.411	0.511	0.92	1

Table 5: Double-peak metrics for retained sparse-testset cases.

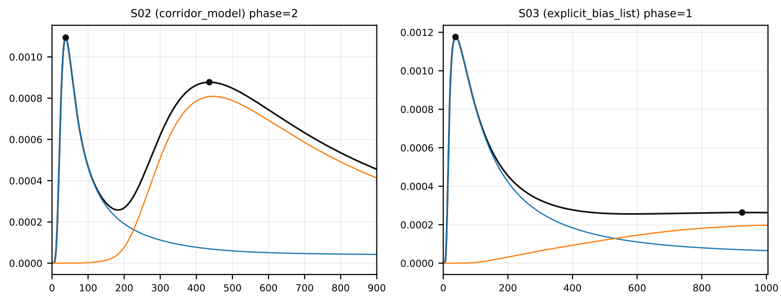

Figure 9: Retained sparse-testset overview (currently S02 clear-double and S03 weak-double).

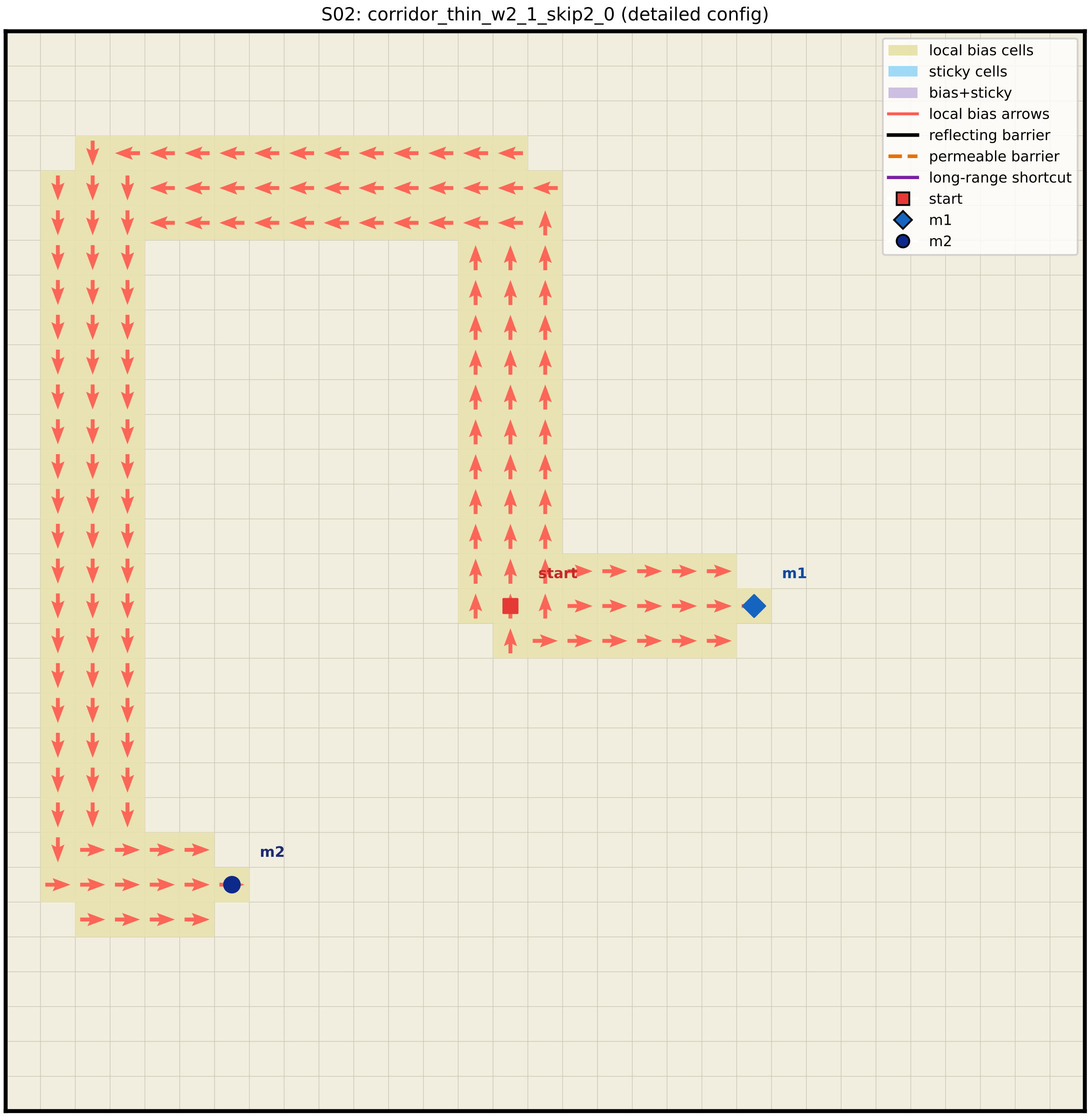

Figure 10: S02 detailed configuration: all biased cells and all arrows shown explicitly.

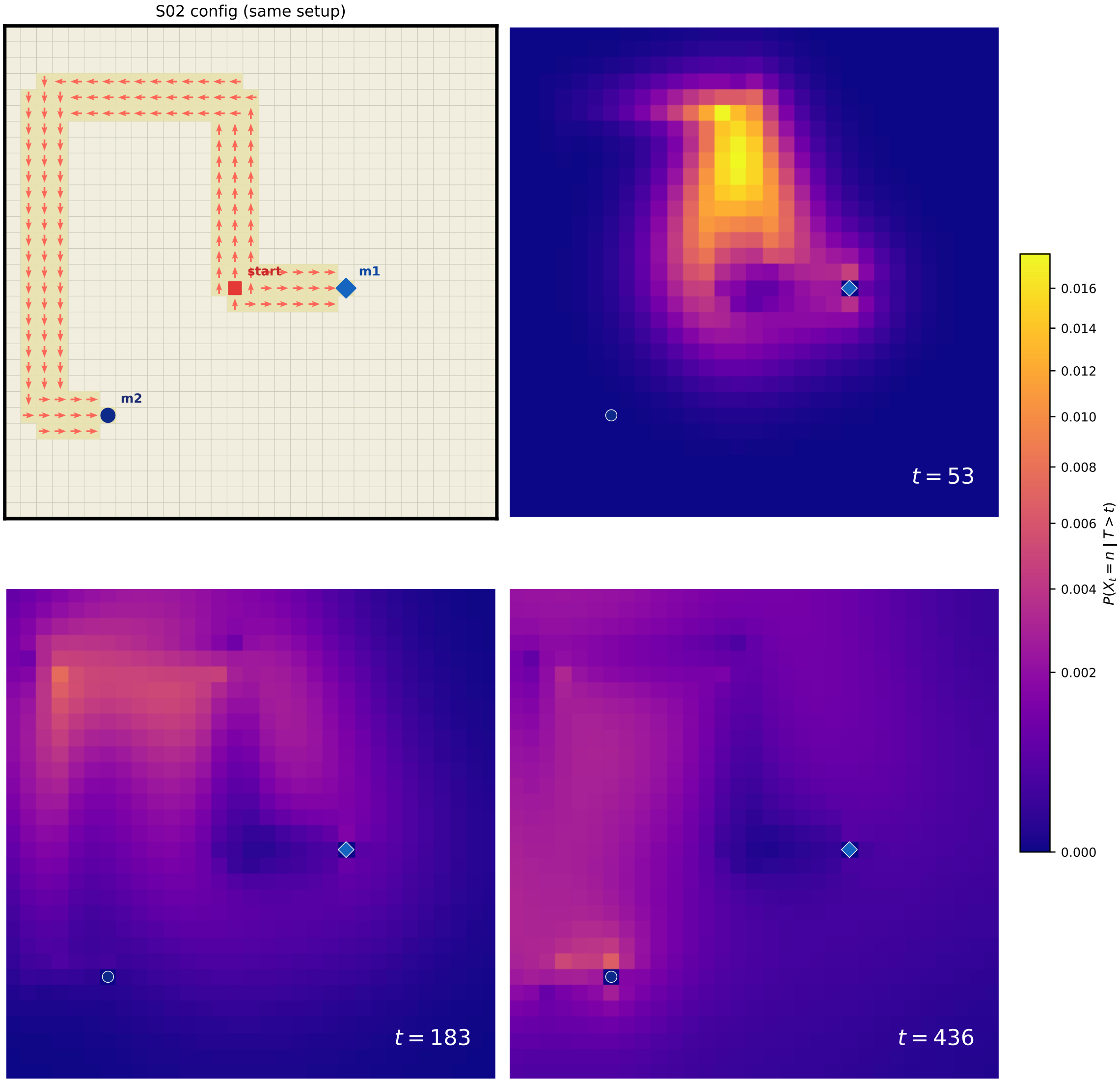

Figure 11: S02 configuration + three-stage conditional occupancy heatmaps (early peak, valley window, late peak window).

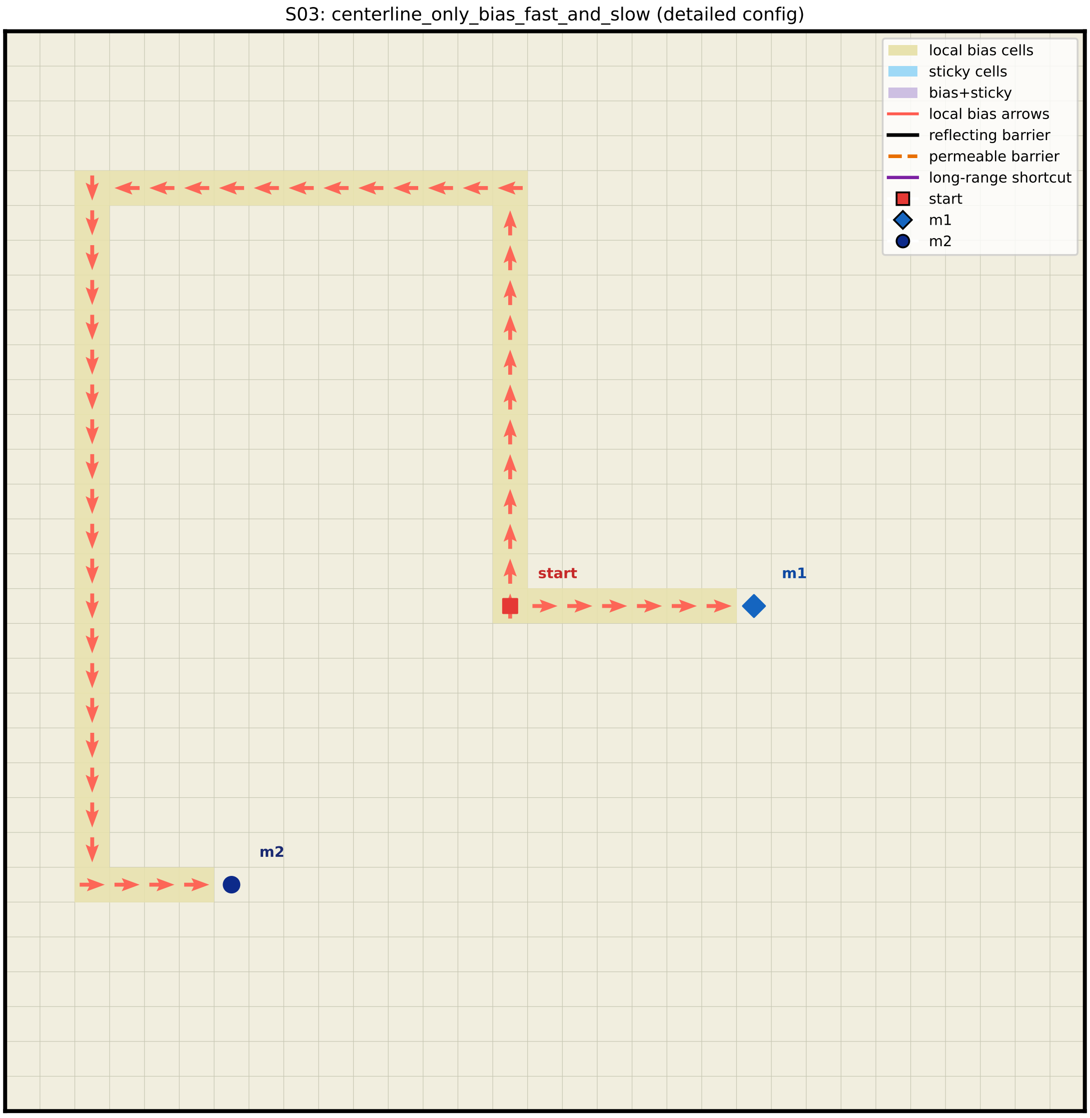

Figure 12: S03 detailed configuration: centerline-only sparse bias layout.

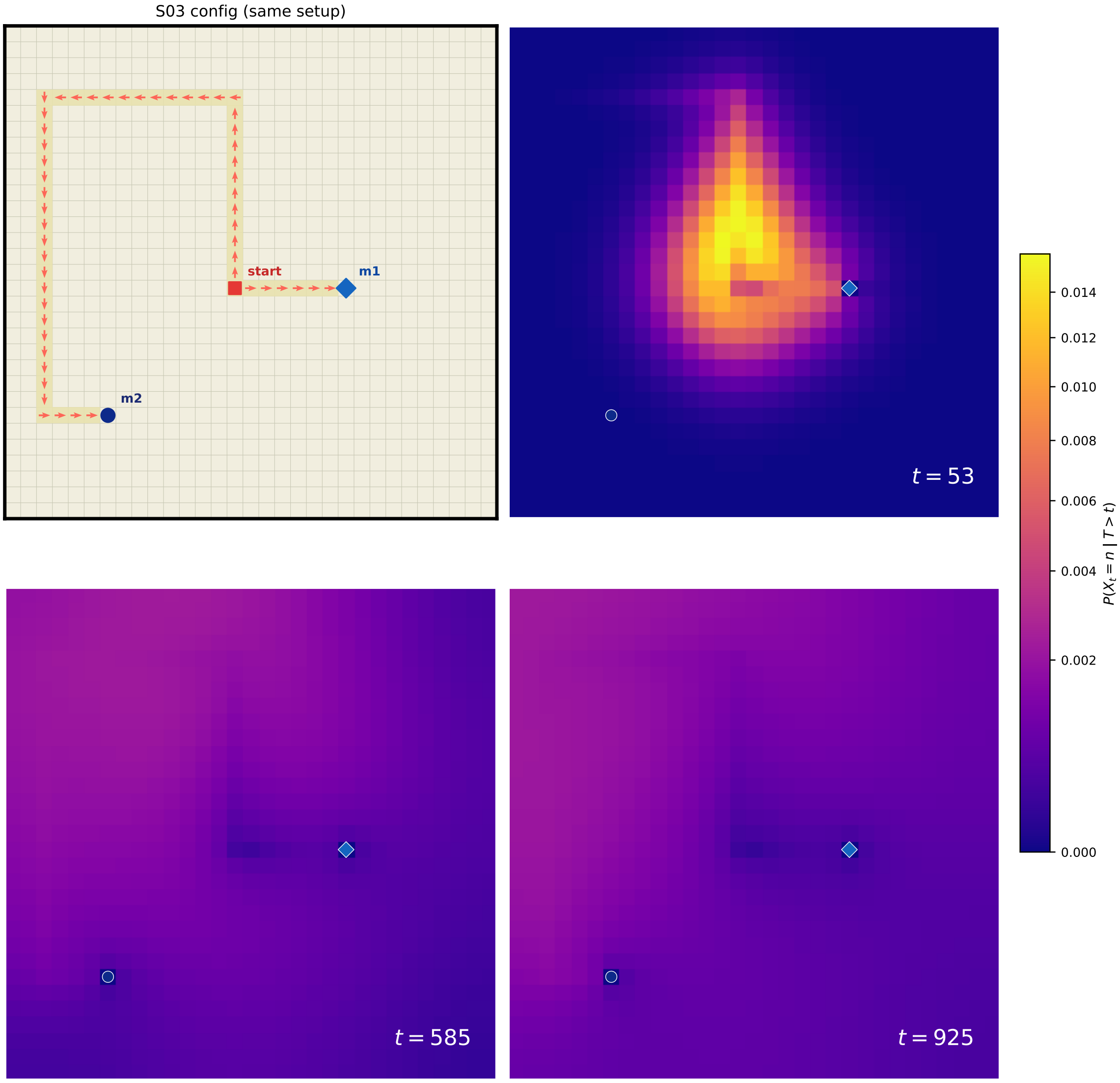

Figure 13: S03 configuration + three-stage conditional occupancy heatmaps (weak-double behavior).

6 Reproducibility

```
python3 scripts/reportctl.py run --report grid2d_two_target_double_peak -- python3 code/two_
python3 scripts/reportctl.py build --report grid2d_two_target_double_peak --lang en
```

Core outputs used by this report:

- `research/reports/grid2d_two_target_double_peak/artifacts/data/case_summary.json`
- `research/reports/grid2d_two_target_double_peak/artifacts/data/sparse_testset_results.json`
- `research/reports/grid2d_two_target_double_peak/artifacts/data/scan_w2_skip2.csv`, `research/reports/grid2d_two_target_double_peak/artifacts/data/scan_w2_skip2.json`
- `research/reports/grid2d_two_target_double_peak/artifacts/outputs/C1_fpt.csv` – `.../C4_fpt.csv`
- `research/reports/grid2d_two_target_double_peak/artifacts/outputs/S02_external_fpt.csv`, `.../S03_external_fpt.csv`

Appendix: Full Arrow-Field Maps (One Arrow per Biased Cell)

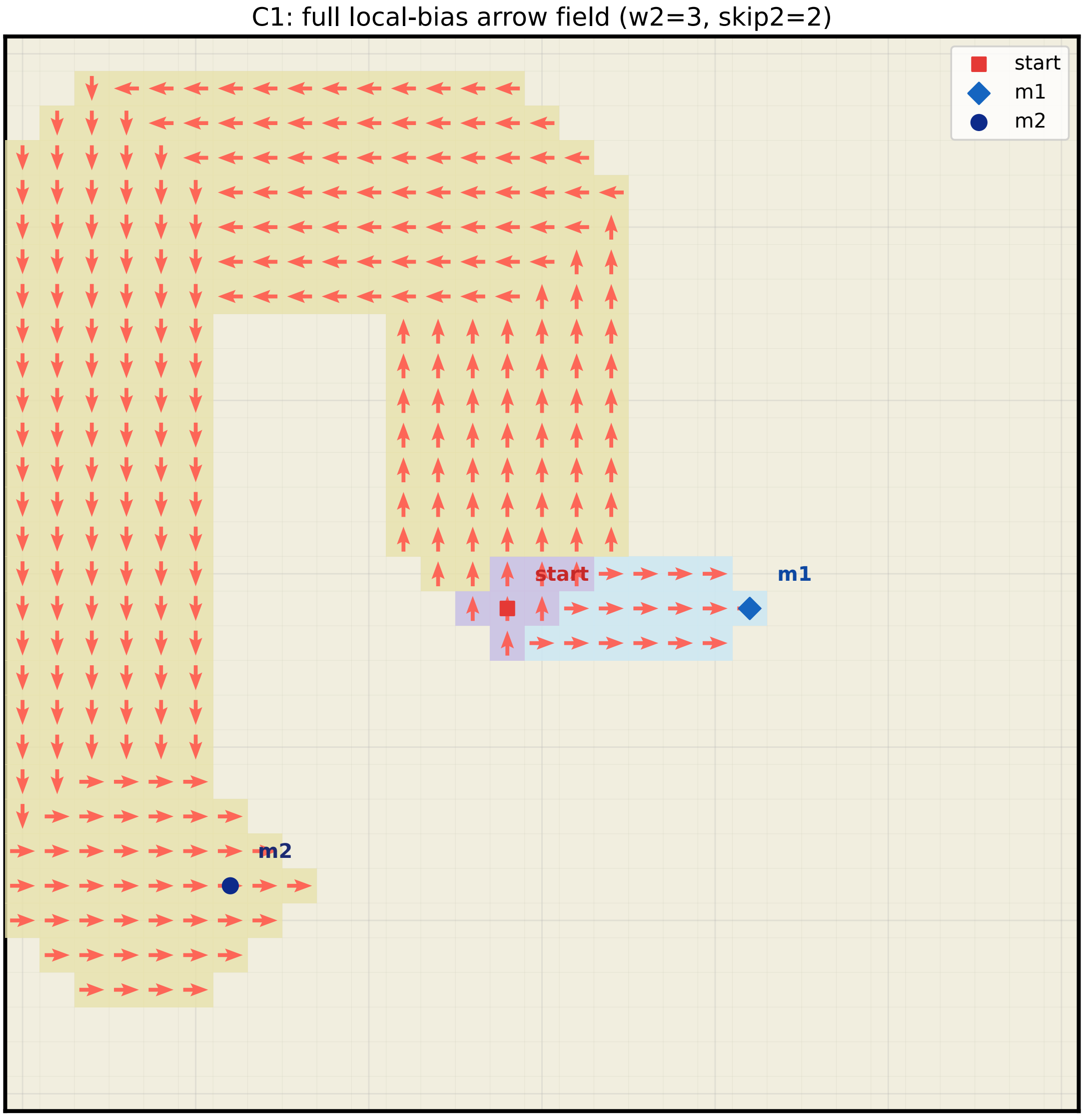

Figure 14: C1 full local-bias arrow field ($w_2 = 3$, $\texttt{skip2} = 2$).

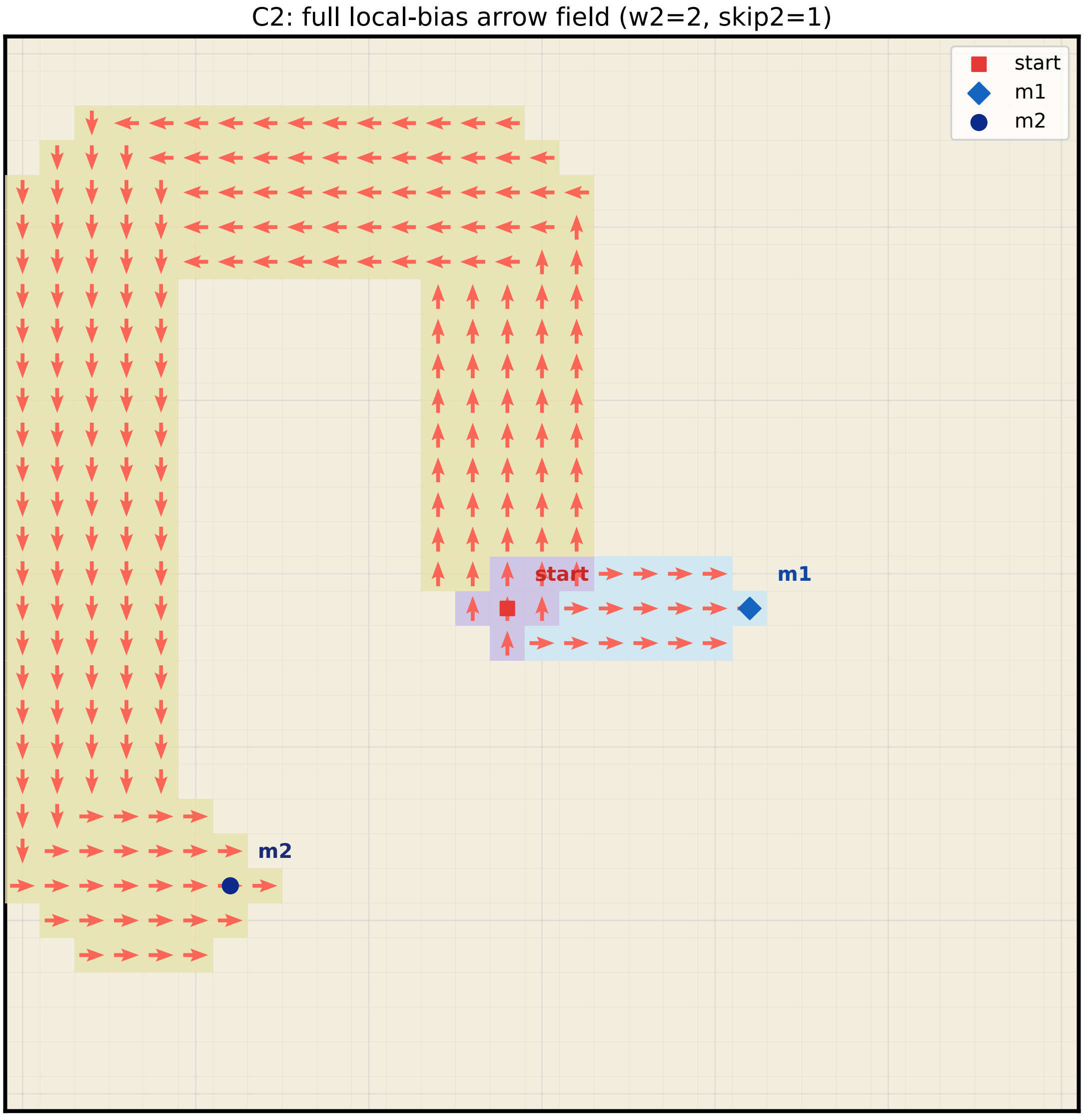

Figure 15: C2 full local-bias arrow field ($w_2 = 2,\ \texttt{skip2} = 1$).

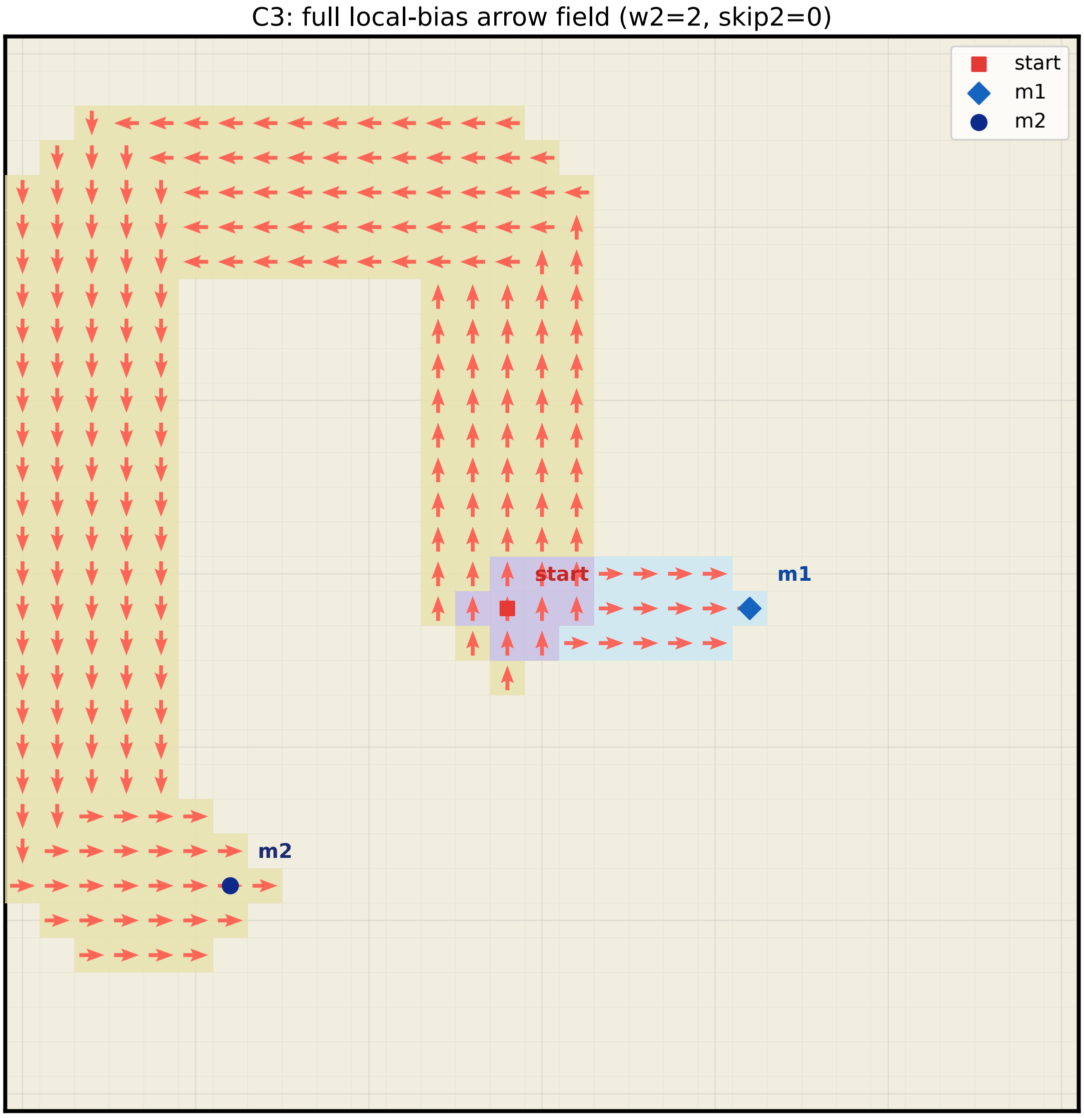

Figure 16: C3 full local-bias arrow field ($w_2 = 2,$ $\texttt{skip2} = 0$).

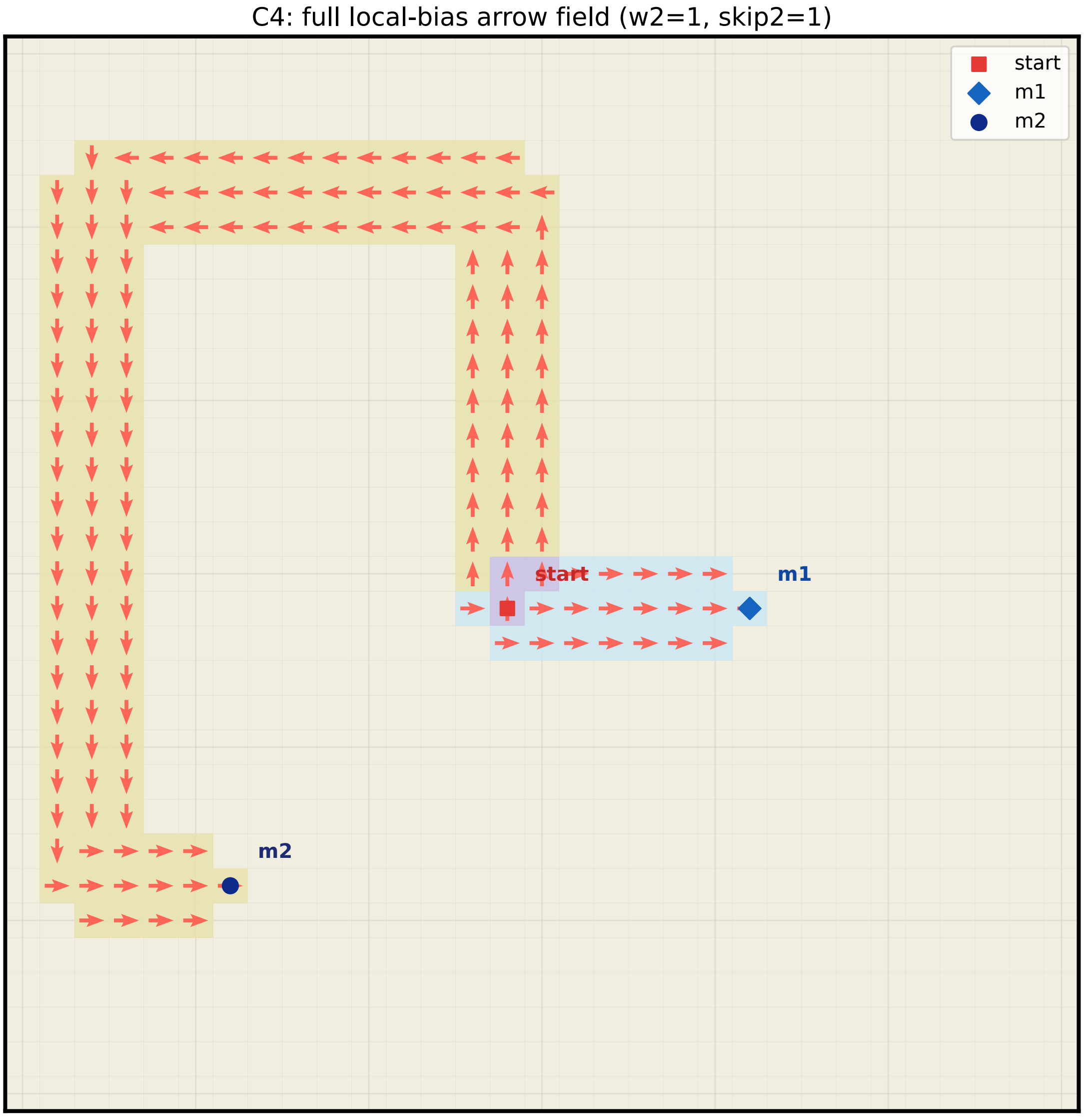

Figure 17: C4 full local-bias arrow field ($w_2 = 1,\ \texttt{skip2} = 1$).